

NOTES TITLE

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Abstract

These notes develop a small piece of theoretical machinery and work through its consequences in detail. The exposition is kept self-contained, with definitions, statements, short proofs, and worked examples interleaved with prose. The closing sections collect a few exercises for practice.

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1 Setting

These notes develop a small piece of theoretical machinery to illustrate the structure of the template. The exposition follows the typical pattern of mathematical and physics notes: definitions, statements of results, short proofs, and worked examples interleaved with prose.

Citations are rendered in brackets, e.g. [1], and multiple sources combine and compress to ranges automatically [1, 2].

2 Definitions and Statements

Definition 2.1. Let $\mathcal{H}$ be a complex separable Hilbert space. A *state* is a unit vector $\psi \in \mathcal{H}$, considered up to a phase factor $e^{i\theta}$.

Definition 2.2. An *observable* is a self-adjoint linear operator $\hat{A}$ acting on $\mathcal{H}$.

The expectation value of an observable in a given state is defined by

$$\langle A \rangle_\psi = \langle \psi | \hat{A} | \psi \rangle. \quad (1)$$

REAL EXPECTATION VALUES

Theorem 2.3. *Expectation values of self-adjoint operators are real.*

Proof. By definition, $\overline{\langle A \rangle_\psi} = \overline{\langle \psi | \hat{A} | \psi \rangle} = \langle \psi | \hat{A}^\dagger | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle = \langle A \rangle_\psi$, where we used $\hat{A}^\dagger = \hat{A}$. $\square$

Remark 2.4. The reality of expectation values is what makes self-adjoint operators the natural mathematical model of observable quantities in quantum mechanics.

2.1 Time Evolution

Time evolution in quantum mechanics is governed by the Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi, \quad (2)$$

where $\hat{H}$ is the Hamiltonian. For time-independent $\hat{H}$, equation (2) integrates to

$$\psi(t) = e^{-i\hat{H}t/\hbar} \psi(0). \quad (3)$$

2.1.1 Stationary states A *stationary state* is an eigenstate of $\hat{H}$; its time dependence is a pure phase, so every expectation value is constant in time. Run-in headings like this one mark the lowest structural level and are kept out of the table of contents.

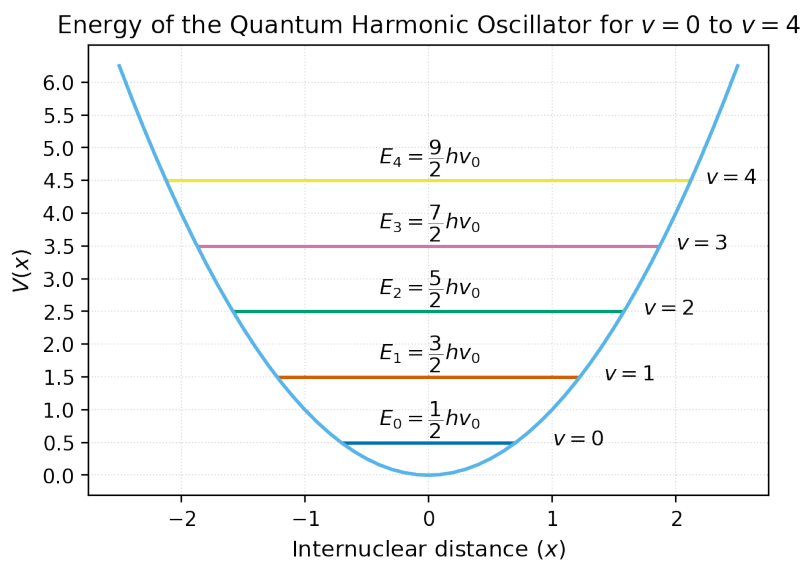


Figure 1. Schematic energy level diagram.

3 Examples

Figure 1 shows a typical structure that recurs throughout the rest of these notes. A few fundamental constants are collected in Table 1 for reference.

Table 1. Fundamental physical constants.

Quantity	Symbol	Value
Speed of light	c	2.998×10^8 m/s
Planck's constant	$\hbar$	1.055×10^{-34} Js
Elementary charge	e	1.602×10^{-19} C

4 Numerical Methods

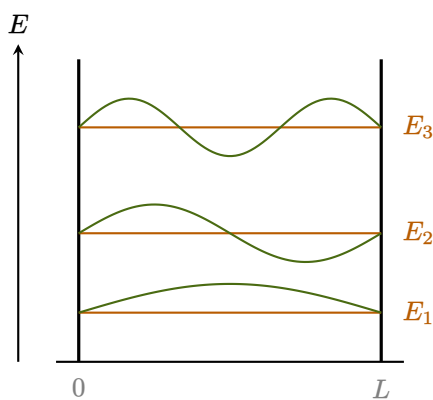


Figure 2. Infinite square well: energy levels $E_n \propto n^2$ (orange) and corresponding eigenfunctions ψ_n (olive).

The `python` environment typesets code with Gruvbox Light syntax highlighting directly in the notes — well suited for short scripts that accompany an analytical derivation. The running example is the infinite square well, shown in Figure 2 with its first three energy levels and eigenfunctions.

The script below evaluates the probability density $|\psi_n(x)|^2$ numerically, then integrates $\langle x \rangle = \int_0^L x |\psi_n|^2 dx$ for the first three levels and confirms the result $\langle x \rangle = L/2$.

```

1  import numpy as np
2
3  def psi(n, x, L=1.0):
4      """Normalized wavefunction for the infinite square well."""
5      return np.sqrt(2.0 / L) * np.sin(n * np.pi * x / L)
6
7  def energy(n, L=1.0, hbar=1.0, m=1.0):
8      """Energy eigenvalue E_n in natural units."""
9      return (n * np.pi * hbar)**2 / (2 * m * L**2)
10
11 x = np.linspace(0, 1, 1000)
12 for n in range(1, 4):
13     prob = psi(n, x)**2          # probability density |psi_n|^2
14     mean_x = np.trapz(x * prob, x)
15     print(f"n={n}   E_n={energy(n):.4f}   <x>={mean_x:.4f}")

```

Running this prints $\langle x \rangle \approx 0.5000$ for all n , consistent with the symmetry of each eigenstate about the midpoint of the well.

5 Exercises

Notes can carry problems alongside the exposition. The `exercise` environment is numbered per section and set at full width, separated from the surrounding text by space alone — less before than after, tying the exercise to its lead-in. `\separator` can additionally mark a change of topic, as below. Subproblems use the `subproblems` environment, which labels parts a), b), c) without affecting ordinary `enumerate` lists.

Exercise 5.1 - Eigenvalues and orthogonality

Show that the eigenvalues of a self-adjoint operator $\hat{A}$ are real, and that eigenvectors belonging to distinct eigenvalues are orthogonal.

Exercise 5.2 - Infinite square well

Consider a particle in a one-dimensional infinite square well of width L .

- a) Write down the normalized energy eigenfunctions.
- b) Compute the energy levels E_n .

The remaining parts build on the spectrum found above; use the result from part (b) when sketching the densities.

- c) Sketch the probability density for $n = 1, 2, 3$.
- d) Compute the expectation value $\langle x \rangle$ in the ground state.

6 Closing

These notes are a template; the content above exists only to demonstrate typography. Replace it with your own material as the notes develop.

References

- [1] Firstname Lastname. Example article title. *Journal Name*, 12(3):45–67, 2025.
- [2] Firstname Lastname. *Example Book Title*. Publisher Name, City, 2025.